CAMPAIGN RESULTS

Track 2H Dispersion-Aware Modeling Campaign Results

Track 2H robust-loss closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Track 2H` robust-loss dispersion-aware campaign. The campaign tested whether less outlier-sensitive pointwise losses help the curve-aware harmonic residual-offset probe handle locally dispersed TE data before moving to quantile, probabilistic, mixture, latent-state, or multi-head architectures.

The campaign completed all planned entries

- `9` completed runs;
- `0` failed runs;
- `3` robust loss profiles;
- `3` required direction surfaces: `global`, `Fw`, and `Bw`.

  The runner-level scalar first entry is `te_track2h_smooth_l1_robust_bw`. This is not a deployment winner by itself. Track 2H must still pass the official curve-first Track 2 verification refresh before promotion decisions.

Campaign Artifacts

Artifact	Path
Campaign output	`output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10`
Leaderboard	`output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10/campaign_leaderboard.yaml`
Best-run pointer	`output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10/campaign_best_run.yaml`
Execution report	`output/training_campaigns/2026-06-11-11-51-10_track2h_dispersion_aware_modeling_campaign_2026_06_10/campaign_execution_report.md`
Planning report	`doc/reports/campaign_plans/track2/2026-06-10-16-56-36_track2h_dispersion_aware_modeling_probe_campaign_plan_report.md`
Campaign package	`config/training/track2h_dispersion_aware_modeling/campaigns/2026-06-10_track2h_dispersion_aware_modeling_campaign`

Execution Summary

Field	Value
Campaign name	`track2h_dispersion_aware_modeling_campaign_2026_06_10`
Completed runs	9
Failed runs	0
Model type	`curve_aware_harmonic_residual_offset_probe`

Field	Value
Loss profiles	`mae` , `smooth_l1` , `log_cosh`
Tested surfaces	`global` , `Fw` , `Bw`
Runner-level scalar first entry	`te_track2h_smooth_l1_robust_bw`
Program-level scalar winner changed	no

Directional Branch Results

Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
`global`	`mae_robust_global`	`mae`	0.003406	0.003807	0.003645
`Fw`	`mae_robust_fw`	`mae`	0.003146	0.003527	0.003258
`Bw`	`smooth_l1_robust_bw`	`smooth_l1`	0.003074	0.003662	0.003372

These rows are the branch-level closeout result. The `Bw` row is also the campaign scalar leader, but it does not replace the `global` or `Fw` branch.

Robust-Loss Leaderboard

Rank	Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
1	`Bw`	`smooth_l1_robust_bw`	`smooth_l1`	0.003074	0.003662	0.003372
2	`Fw`	`mae_robust_fw`	`mae`	0.003146	0.003527	0.003258
3	`Fw`	`smooth_l1_robust_fw`	`smooth_l1`	0.003314	0.003679	0.003235
4	`Fw`	`log_cosh_robust_fw`	`log_cosh`	0.003355	0.003708	0.003280
5	`global`	`mae_robust_global`	`mae`	0.003406	0.003807	0.003645
6	`global`	`smooth_l1_robust_global`	`smooth_l1`	0.003422	0.003810	0.003641
7	`Bw`	`mae_robust_bw`	`mae`	0.003430	0.004029	0.003579
8	`Bw`	`log_cosh_robust_bw`	`log_cosh`	0.003481	0.004029	0.003774
9	`global`	`log_cosh_robust_global`	`log_cosh`	0.003505	0.003935	0.003645

Baseline Comparison

Surface	T2H Best	T2H MAE	T2G Ref.	Ref. MAE	Delta
`global`	`mae_robust_global`	0.003406	`full_curve_composite_global`	0.003345	-1.82%
`Fw`	`mae_robust_fw`	0.003146	`raw_centered_shape_fw`	0.003181	+1.10%
`Bw`	`smooth_l1_robust_bw`	0.003074	`pointwise_control_bw`	0.003430	+10.38%

Scalar training metrics show a useful but direction-dependent result. Robust losses clearly improve the backward scalar branch relative to the Track 2G backward reference and slightly improve the forward scalar branch. The global branch does not improve over the Track 2G full-curve-composite reference.

Loss Interpretation

Loss Profile	Observed Signal	Interpretation
`mae`	Best `global` and best `Fw` Track 2H scalar result.	Absolute-error pressure is useful where local dispersion likely distorts MSE-style training.
`smooth_l1`	Best `Bw` and campaign scalar leader.	Huber-like behavior is the strongest first signal for the backward dispersion/offset problem.
`log_cosh`	No branch winner and weakest global result.	It is stable but not competitive in this first robust-loss campaign.

The main modeling result is not that robust losses solve Track 2. The useful result is that robust loss selection matters, especially on the backward surface. This supports continuing the dispersion-aware plan before freezing a multi-head architecture.

Registry Effects

The campaign runner refreshed family registries for all nine Track 2H families and updated the program registry. The program-level scalar best did not move: `te_periodic_gru_sequence_remote_Bw` remains the current scalar program winner with test `MAE = 0.002344`.

Registry Scope	BestRun	Test MAE
`Track 2H global robust`	`te_track2h_mae_robust_global`	0.003406
`Track 2H Fw robust`	`te_track2h_mae_robust_fw`	0.003146
`Track 2H Bw robust`	`te_track2h_smooth_l1_robust_bw`	0.003074

Track 2 Boundary

Official Track 2 curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

The next Track 2 package should add all nine Track 2H candidates and report accepted results separately for:

- `global`;
- `Fw`;
- `Bw`.

  The verification must compare raw curve error, centered-shape error, offset, amplitude, harmonic behavior, collage plots, and overlay plots against Track 2G, Track 2F-bis, Wave 2B, and the current accepted Track 2 baselines.

Closeout Decision

Track 2H robust-loss execution is complete: all planned candidates have successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry `mae_robust_global` forward as the strongest Track 2H global scalar candidate;
- carry `mae_robust_fw` forward as the strongest Track 2H forward scalar candidate;
- carry `smooth_l1_robust_bw` forward as the strongest Track 2H backward scalar candidate;
- keep `log_cosh` as a completed negative/weak baseline, not the next default loss profile;
- do not promote Track 2H before official Track 2 verification.

Recommended Follow-Up

1. Accept this closeout and clear the active campaign state.
2. Prepare a separate operator-launched Track 2 verification refresh for all nine Track 2H candidates.
3. If the official Track 2 refresh confirms the scalar backward improvement, make robust losses part of the candidate set for later multi-task / multi-head models.
4. Prepare the next Track 2H package in staged order: quantile/probabilistic regression, mixture-density heads, then latent-state or hysteresis-aware models.
5. Keep `Fw`, `Bw`, and `global` as parallel branch decisions.